

H1 SF2524

Anh Do, Tommaso Brambilla

anhd@kth.se , brambi@kth.se

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Exercise 1

1.a

Here we implemented the power method and from the graph we can see that the convergence of the eigenvalue error $|\lambda^{(k)} - \lambda_1|$ is following the theoretical convergence rate of $O\left(\left(\frac{|\lambda_2|}{|\lambda_1|}\right)^{2k}\right)$, where λ_1 is the largest absolute eigenvalue of the matrix A and λ_2 is the second largest absolute eigenvalue, k is the index of the iteration.

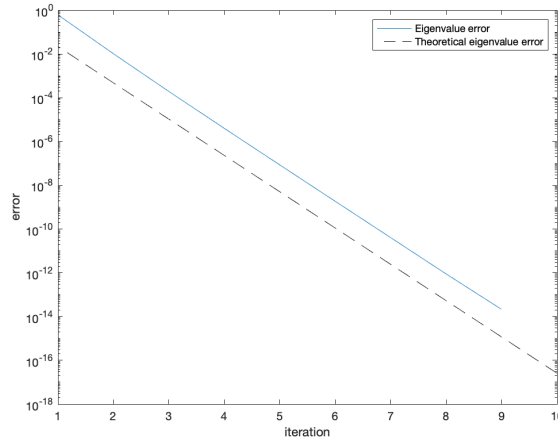


Figure 1: Semi logarithmic plot of the eigenvalue error and its convergence rate for the power method

1.b

Here we implemented the Rayleigh quotient iteration and we analysed the convergence of the eigenvalue error (for the implementation of the method, we chose $\mu = \frac{x'Ax}{x'x}$ at the first iteration. The eigenvalue error at the iteration k was defined as $|\lambda^{(k)} - \lambda|$, where λ is the final approximation, this is because we don't really know to which eigenvalue of A the RQI will converge, so the last iteration is the most reasonable value to take as reference).

In the graph below the eigenvalue error is compared with the functions ϵ (Reference 1), ϵ^2 (Reference 2), ϵ^3 (Reference 3), where ϵ is the error of the eigenvector. It is clear how the eigenvalue error (blu line) follows the convergence of ϵ^2 , as it should be since A is symmetric.

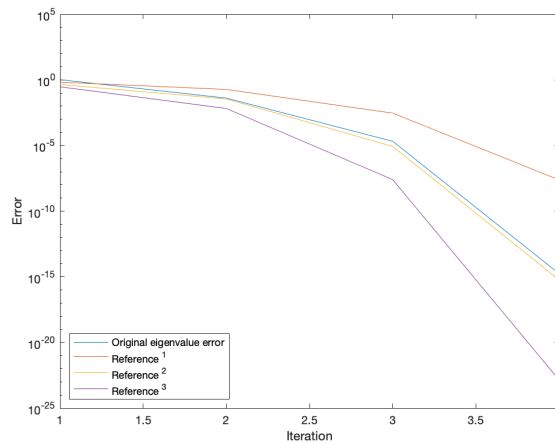


Figure 2: Semi logarithmic plot of the eigenvalue error and its convergence rate for the Rayleigh quotient iteration

1.c

The convergence of the Rayleigh Quotient Iteration is slower when changing the element $a_{1,3}$ to 4, because the A matrix is no more symmetric and the Rayleigh quotient accuracy changes from $\mathcal{O}(\epsilon^2)$ to $\mathcal{O}(\epsilon)$

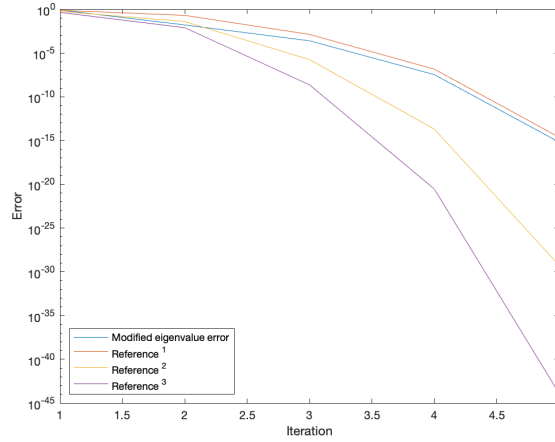


Figure 3: Semi logarithmic plot of the eigenvalue error and its convergence rate for the Rayleigh quotient iteration but with an unsymmetric matrix

Exercise 2

2.a

	single GS		modified GS		double GS		triple GS	
	time	orth	time	orth	time	orth	time	orth
$m = 5$	0.731169	3.9879e-13	0.708851	3.9879e-13	0.486280	9.7917e-14	0.415080	1.000e-13
$m = 10$	1.657926	3.9879e-13	1.683719	3.9879e-13	1.252394	1.0202e-13	1.057311	1.003e-13
$m = 20$	4.272772	1.4068e-13	4.166520	5.7906e-14	3.551333	8.9858e-14	3.244074	2.4883e-14
$m = 50$	17.380391	1.0349e-12	17.115119	4.5778e-13	16.851038	4.9644e-13	16.081235	1.2598e-13
$m = 100$	56.246460	3.8425e-12	55.848073	2.0351e-12	59.777176	1.5851e-12	58.430678	4.3691e-13

Table 1: Comparison of GS methods

2.b

The classical GS is really sensitive to round-off errors in certain situations, more precisely, the Classical GS method has a large round-off error if the vector b (starting vector) almost lies in the subspace $\text{span}(Q)$.

The modified GS procedure is equivalent to the CGS in exact arithmetic, but different in floating-point arithmetic (actually this does not imply that MGS is always better in floating-point arithmetic than CGS).

For the repeated GS procedures (double and triple), the idea is to redo the orthogonalization more than once since after the CG, it won't really be orthogonal due to round-off errors.

In this exercise, the value of **orth** is quite always increasing with the number of iterations **m**, this is probably due to the higher repetition of rounding errors when increasing the iterations. It is important to notice that, the Double and Triple GS give more accurate results (their values of **orth** is smaller than the ones in Classical and Modified GS for almost all the values of **m**).

The obtained results seem to be reasonable compared to the theory: the repeated GS methods are more efficient and it is not clear if the Modified GS gives better results than the classical one, since there is not much difference both in the **time** and **orth** columns.

Exercise 3

3.a

For this proof, we use two lemmas presented in previous lectures:

Lemma 1:

If Q_m and H_m satisfy the Arnoldi factorisation and $K_m(A, b) = [b, Ab, \dots, A^{m-1}b]$ has full column rank, then $\text{span}(K_m(A, b)) = \text{span}(Q_m)$.

Lemma 2:

If Q_m and H_m satisfy the Arnoldi factorisation, then $H_m = Q_m' A Q_m$.

Proof:

The eigenvalue approximations provided by the Arnoldi method are the eigenvalues of H_m . We want to show that the matrices H_m and $(K_m' K_m)^{-1} K_m' A K_m$ have the same eigenvalues. Since Q_m is the result of the orthogonalisation of K_m , there exists an invertible matrix S such that:

$$K_m S = Q_m.$$

From Lemma 2, we have:

$$H_m = Q_m' A Q_m.$$

Substituting $K_m S = Q_m$ into the expression for H_m , we get:

$$H_m = (K_m S)' A K_m S = S' K_m' A K_m S.$$

Assume:

$$H_m w = \mu w \quad \Longleftrightarrow \quad S' K_m' A K_m S w = \mu w.$$

Multiplying both sides by $(S')^{-1}$ on the left yields:

$$K_m' A K_m S w = \mu (S')^{-1} w.$$

Define a new vector $u = S w$, which implies $w = S^{-1} u$. We can now rewrite the equation as:

$$K_m' A K_m u = \mu (S')^{-1} S^{-1} u.$$

Since:

$$(S')^{-1} S^{-1} = (K_m' K_m)^{-1}$$

(because $I = Q_m' Q_m = S' K_m' K_m S$), we have:

$$K_m' A K_m u = \mu (K_m' K_m)^{-1} u.$$

This leads to:

$$(K_m' K_m)^{-1} K_m' A K_m u = \mu u.$$

Thus, we have shown that H_m and $(K_m' K_m)^{-1} K_m' A K_m$ have the same eigenvalues.

3.b

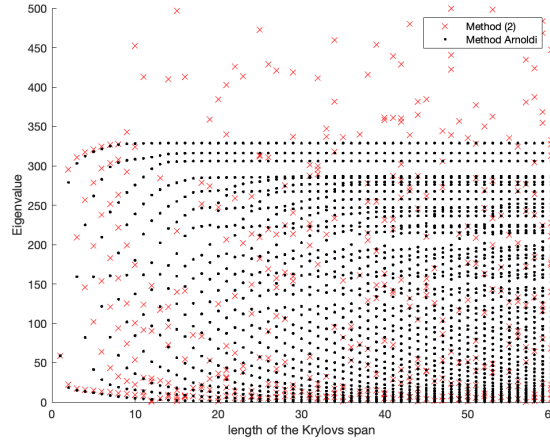


Figure 4: Plot of the eigenvalues generated by the Arnoldi method and the (2) method from 3a

3.c

Interpreting the results from **3.a**, we observe that the two algorithms are the same in exact arithmetic, in the proof there are neither rounding nor approximations of any kind, so the eigenvalues approximations given by the Arnoldi method and the method without orthogonalization, should be exactly the same. However, in **3.b** we observe a different result, in particular it is clear that the Arnoldi method has a better stability while some of the eigenvalues provided by the second method seem to diverge from the main pattern. This issue comes from the rounding errors, the main reason of doing orthogonalization of the Krylov subspace in the Arnoldi method is to avoid rounding errors.

Exercise 4

4.a

From the theory about the convergence of the Arnoldi method, we know that the most extreme and isolated eigenvalues will be the ones having the fastest convergence. That is because the quantity $\|(I - QQ^*)x_i\|$ (we take this quantity to measure the convergence towards the eigenpair (x_i, λ_i)) is bounded by the product of the quantities $\epsilon_i^{(m)}$ and ζ_i , and $\epsilon_i^{(m)}$ is in turn bounded by $(\frac{\rho}{|\lambda_i - c|})^{m-1}$, where ρ is the radius of the smallest circle including all the eigenvalues except λ_i and c is the center of this circle.

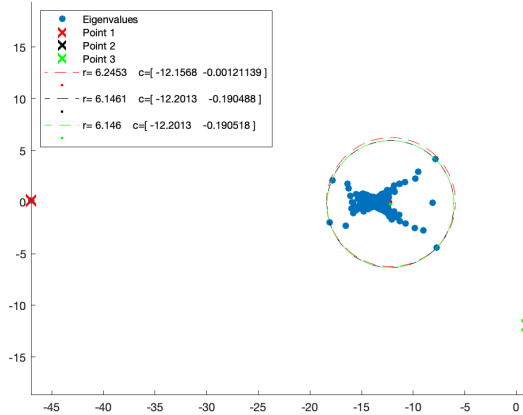


Figure 5: Plot of the eigenvalues generated by the Arnoldi method with extra references for calculating the convergence factor

4.b

Here we are considering the same circle to compute the convergence factor for all the three isolated eigenvalues (this circle, as shown in the graph, does not include the three isolated eigenvalues, we are justified in doing so since having two outliers does not change much the convergence factor). We can calculate the convergence factor with the formula $\frac{\rho}{|\lambda_i - c|}$ for the three different points (λ_r is the red point, λ_g is the green point and λ_b is the black point):

- $\frac{\rho}{|\lambda_r - c|} = 0.1791$
- $\frac{\rho}{|\lambda_g - c|} = 0.3295$
- $\frac{\rho}{|\lambda_b - c|} = 0.3485$

4.c

The following figure clearly illustrate that we converge relatively fast to the three points λ_r , λ_g and λ_b

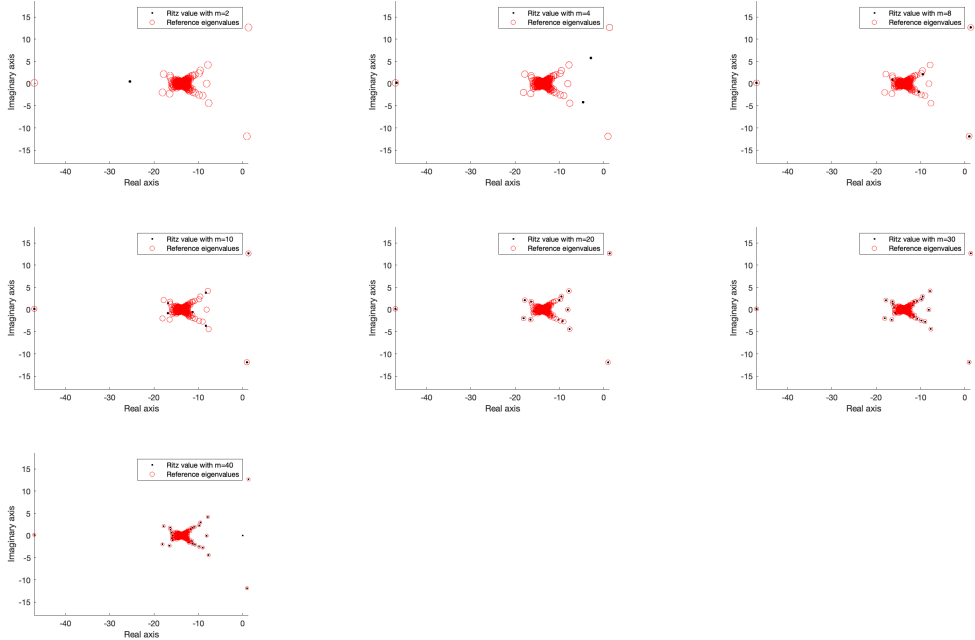
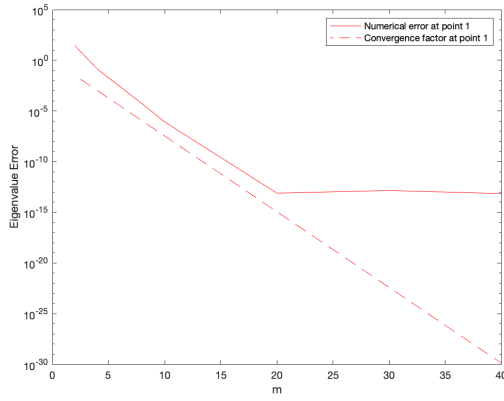
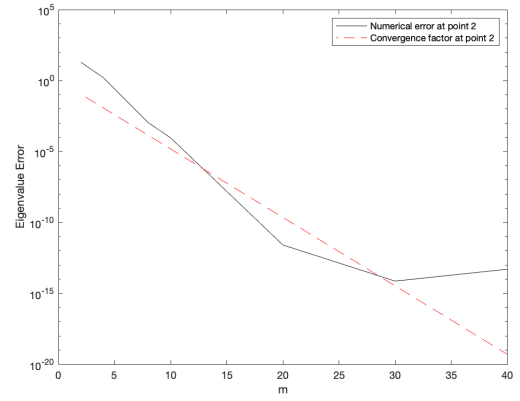


Figure 6: Plot of the eigenvalues generated by the Arnoldi method with different m

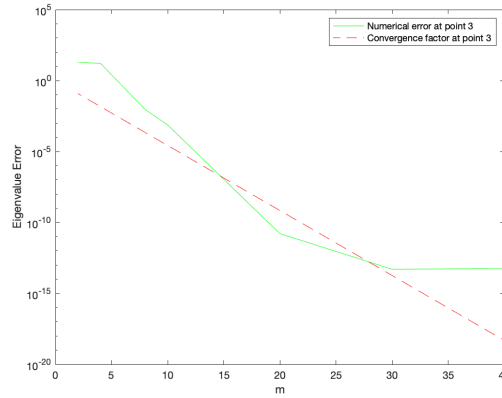
Based on the figure 7 bellow, we can see that the eigenvalue error is less than 10^{-10} when $m \approx 15$ for λ_r , $m \approx 18$ for λ_g and $m \approx 18$ for λ_b . We can also see from figure 7 that for the plot (a) the theory seems to overlap well with the results that we get. It is worth mentioning that the convergence factor is an upper bound for our error which is not well illustrated by the figures as we have not taken into consideration the ζ_i factor. We therefore have to assume that the ζ_i is large enough such that the dashed line would cover our result line. This explanation holds well for the plot (b) and (c) as the theory's slope is greater than the slope of the results line at least before we approach floating point error.



(a) Error plot for λ_r



(b) Error plot for λ_b



(c) Error plot for λ_g

Figure 7: Error plots for different λ_i with respect to the Krylov span size determined by m .

4.d

The shift-and-invert Arnoldi method often has faster convergence towards the eigenvalues closest to the shift, and we can verify this from the table below: when $\sigma = -9.8 + 1.5i$ (really close to $-9.8 + 2i$) we have the fastest convergence (since we have the smallest error with the same number of iterations). For the same reason we have faster convergence when $\sigma = -7 + 2i$ than when $\sigma = -10$. The Arnoldi method has quite a big error, especially when m is small since the eigenvalue $-9.8 + 2i$ is placed in the middle of the group of eigenvalues, not isolated at all.

	Standard AM	$\sigma = -10$	$\sigma = -7 + 2i$	$\sigma = -9.8 + 1.5i$
$m = 10$	2.5	0.0092146	0.0087514	7.3803e-06
$m = 20$	0.6	0.00025317	9.108e-08	6.8245e-14
$m = 30$	$2 \cdot 10^{-4}$	3.0717e-08	1.0367e-13	5.575e-14

Table 2: Error values for different shifts and m values

Exercise 5

5.a

In the video it is explained how the large sparse eigenvalue problems arise in the analysis of data, in particular the graph-based method is explained, a method belonging to the field of exploratory analytics. Given a big amount of data, the graph method consists of mapping the data into a graph, where every node corresponds to a data point and the edges describe the similarity between two data points. The graph is then associated with the Laplacian matrix, where each element is different from zero only if there is similarity between the two elements in the respective row and column. Since in big data most data points are not similar, the Laplacian

matrix is often large and sparse. Finding the eigenpairs of the matrix becomes important because eigenvectors provide insight into data subgroups.

5.b

The Rayleigh-Ritz method is a procedure that, given an orthogonal matrix $Q \in R^{n \times m}$ where $n > m$, finds the approximations for the eigenpairs of a matrix $A \in R^{n \times n}$. The assumption at the base of the method is that the eigenvectors of A can be well approximated from vectors lying in the subspace formed by the span of the columns of Q . The eigenpairs approximations are the eigenpairs of the matrix $H = Q' A Q \in R^{m \times m}$, which are less expensive to compute since H has smaller dimensions than A . In particular, if the eigenvectors of A lie in the subspace, then the Rayleigh-Ritz method is correct. It is convenient to use this method instead of computing the actual eigenpairs of A , when $m \ll n$, because the matrix $H = Q' A Q$ is $m \times m$ and computing its eigenpairs is much less expensive.

5.c

The breakdown is the situation in which the Arnoldi method fails, it happens when beta is equal to zero at a certain iteration. Actually, having breakdown is equivalent to have an optimal situation, in particular it is equivalent to have the coefficient $h_{m+1,m} = 0$ and to have exact Ritz values. These implications are justified by the following **Lemma**:

If Q_{m+1} and H_m satisfy the Arnoldi factorization and $\mu z = H_m z$, then, with $v = Q_m z$, $\|A - \mu v\| = |h_{m+1,m}| e'_m z$, where e_m is the m th unit vector

Proof:

$$Av - \mu v = A Q_m z - Q_m \mu z$$

Since $\mu z = H_m z$ and using the Arnoldi factorization in the following form: $A Q_m = Q_m H_m + h_{m+1,m} q_{m+1} e'_m$ we obtain:

$$Av - \mu v = Q_m H_m + h_{m+1,m} q_{m+1} e'_m - Q_m H_m = h_{m+1,m} q_{m+1} e'_m$$

Taking the norm of both left and right terms of the equation, since $\|q_{m+1}\| = 1$ the statement of the Lemma is proved.

5.d

The relation between the Arnoldi method and the Shift-and-invert Arnoldi method is similar to the one between the power method and the inverse iteration. In particular, if with the Arnoldi there is fast convergence towards the extreme and isolated eigenvalues, with the Shift-and-invert Arnoldi there is often fast convergence to the eigenvalues close to the shift μ . The main idea is to apply the AM to the matrix $B = (A - \mu I)^{-1}$. In order to make the algorithm faster and avoid to compute the inversion of the shifted matrix $(A - \mu I)$, it is recommended to precompute a factorization (e.g. LU -factorization). Then, instead of computing the eigenvalues of the Hessenberg matrix H , as in AM, here the approximations are provided by the values $1/(\text{eigvals}(H) - \mu)$